Replacing the Linear Local Model of Interpretable Mesomorphic Networks with a Tsetlin Machine

IMN × TM research programme, `experiments/IMN`

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Abstract

Interpretable Mesomorphic Networks (IMN; Kadra et al., NeurIPS 2024) train a deep hypernetwork that writes, for every example, a linear model in the original feature space; the linear model is both the prediction and its explanation. Their stated limitation is that the per-example models are linear. We replace the linear model by a *Tsetlin machine* (TM), a weighted vote of conjunctive rules over thresholded features. Because TM clauses are learned by discrete feedback and have no gradient, we train a TM first, freeze its clauses, and let the hypernetwork generate per-example *clause weights* (IMN-TM); we also study a linear+rules hybrid and a model in which the TM itself plays the hypernetwork (TM-Gated, no black box).

Accuracy. On 12 AutoML-benchmark datasets with default hyper-parameters (720 runs), IMN-TM improves on its source TM on 11/12 datasets (Holm $p = 0.024$) and ties IMN on average (ΔAUROC +0.0006). The per-dataset difference tracks how much tree ensembles beat linear models (Pearson $r = 0.92$). After a 20-trial search per method (720 final runs), IMN-TM beats its TM on 12/12 datasets and a global re-fit of its clause weights (Holm $p = 0.031$). It shares the best average rank with gradient boosting (3.75), ahead of IMN (7.25). Its margin over IMN (+0.0058) is not significant, and the tree-gap pattern disappears because tuning mostly helps IMN on threshold-shaped data.

Explanations. Clause explanations remain valid on an example's neighbourhood: local fidelity is 0.72 vs. 0.57 for IMN. On 60 synthetic tasks with known contributions, they are closer to the truth than IMN's on 41 tasks (Wilcoxon $p = 2.4 \times 10^{-4}$). The price is length: 51–126 weighted rules vs. 15 features cover 90% of an explanation.

Larger datasets. On 20 larger datasets under a fixed training budget, IMN-TM beats IMN significantly (+0.0183, Holm $p = 0.030$). The gain comes almost entirely from datasets where the step cap leaves IMN under-trained ($r = -0.88$ with IMN's shortfall against the paper); elsewhere it is +0.0027.

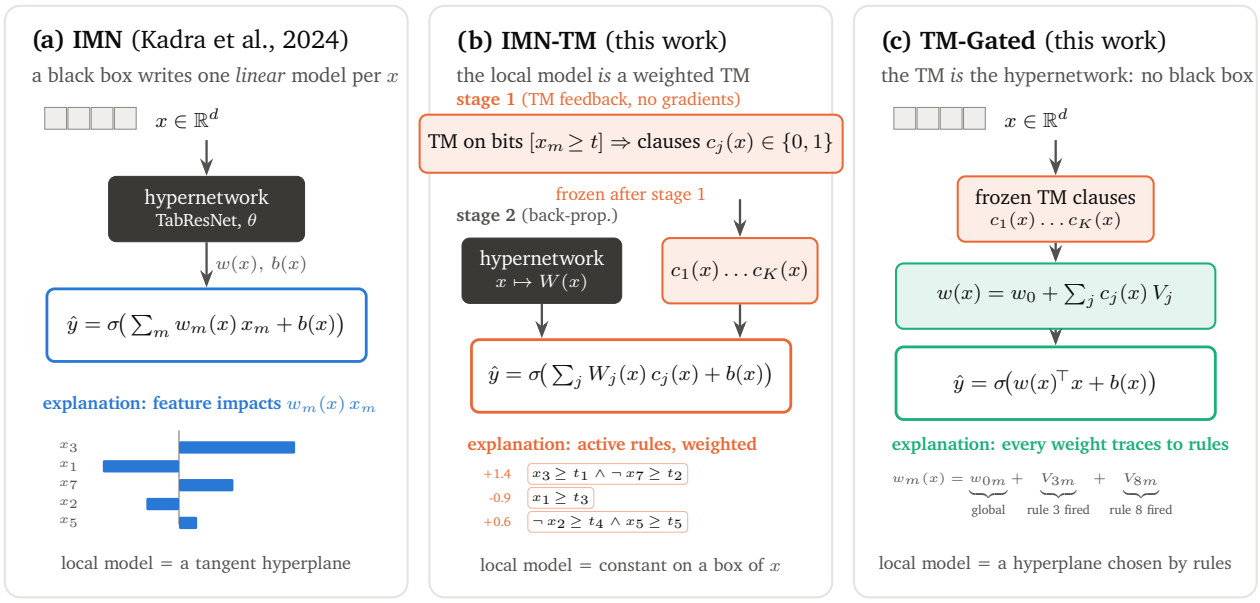
What matters. Ablations show that learned clauses matter for explanation fidelity, not for accuracy, and that initialising at the TM's votes matters for fidelity and for budget-limited training.

1 Introduction

IMN [1] make a deep tabular model explainable by construction: a hypernetwork $w(x;\theta)$ outputs the parameters of a linear model, and the logit is that linear model evaluated at x, $z_k(x) = \sum_m w_{km}(x)\, x_m + w_{k0}(x)$. The explanation of x is the vector of feature impacts $w_m(x)\, x_m$. Section 7 of the paper names the obvious limitation: the local models are hyperplanes, and it suggests non-linear interpretable local models as future work.

A Tsetlin machine [2] is an interpretable non-linear model of exactly that kind: a weighted sum of conjunctions (*clauses*) of Boolean literals, $v_k(x) = \sum_j W_{kj}\, c_j(x)$, where on tabular data each literal is a threshold test $x_m \geq t$ or its negation. This report asks whether the linear local model of IMN can be replaced by a TM, what that costs in accuracy, and what it changes about the explanations. Contributions:

1. a feasibility analysis and three constructions (Section 2), implemented on top of the `torchtsetlin` library without modifying it;
2. a faithful re-implementation of IMN (read from the official code) and a controlled comparison with the original TM, IMN and the paper's baselines on the AutoML benchmark, with default and equal-budget tuned hyper-parameters;
3. interpretability measurements that apply to every method: removal attributions, explanation size, local fidelity, self-consistency, and ground-truth attribution quality on synthetic tasks.



What we found (12 AutoML-benchmark datasets × 5 seeds, tuned; 60 synthetic tasks)

Better than the TM. IMN-TM beats its own source TM on 12/12 datasets (mean ΔAUROC +0.0094, Holm $p = 0.005$).

Level with the best. Tuned, IMN-TM shares the best average rank with GBDT (3.75; IMN 7.25); vs. IMN: +0.0058, not significant.

More faithful explanations. Neighbour accuracy 0.72 vs. 0.57 for IMN; closer to the true attributions on 41/60 synthetic tasks.

The price is length. 51 (default) to 126 (tuned) weighted rules cover 90% of an IMN-TM explanation, vs. 15 features for IMN.

Figure 1: Graphical abstract. (a) IMN: a TabResNet hypernetwork generates a linear model per example. (b) IMN-TM: a Tsetlin machine is trained first (stage 1) and its clauses are frozen; the hypernetwork then generates one weight per clause and example (stage 2), so the local model is a weighted TM. (c) TM-Gated: the frozen clauses themselves generate the coefficients of a linear model, so no black box remains. Rules in (b) and coefficients in (c) are schematic; Figure 6 shows a real explanation. Findings: tuned hyper-parameters, 12 datasets × 5 seeds, and the synthetic benchmark of Section 4.6.

2 Method

Readers who prefer numbers to symbols can start with Section 2.4. It follows one toy example through every equation of this section.

2.1 Why the clauses must be frozen

A hypernetwork is trained by back-propagation, so whatever it generates must enter the loss differentiably. A TM clause is a discrete set of included literals learned by Type I/II feedback; it has no gradient. The clause *weights*, however, are continuous, and once the clauses are fixed their outputs $c_j(x) \in \{0, 1\}$ are ordinary Boolean features of x. This gives three constructions (Figure 1); a fourth—a hypernetwork that emits the clauses themselves—is not pursued: the include masks are discrete, and a rule generated *for* x and evaluated *at* x is trivially true or false, so it explains nothing unless constrained to a neighbourhood.

IMN-TM (local model = weighted TM). With a frozen clause basis $c(x) \in \{0, 1\}^K$ from a TM trained in stage 1, the hypernetwork outputs $W(x) \in \mathbb{R}^{K \times n_{\text{out}}}$ and $b(x)$, and

$$z_k(x) = \sum_{j=1}^{K} W_{kj}(x)\, c_j(x) + b_k(x). \tag{1}$$

The output layer is zero-initialised and its bias set to the TM's signed clause votes divided by T, so before training IMN-TM *is* the source TM (unclamped), for every input. The explanation is the list of active rules with their weights $W_{kj}(x)$.

IMN-Hybrid. The local model has both terms, $z_k(x) = \sum_m w_{km}(x) x_m + \sum_j W_{kj}(x) c_j(x) + b_k(x)$.

TM-Gated (TM as the hypernetwork). The local model stays linear, but its coefficients are a linear function of the clause outputs, $w_k(x) = w_{0k} + \sum_j c_j(x)\, V_{jk}$, learned by gradient descent with the clauses frozen. Every coefficient of every local model is a global term plus one term per fired rule, so the whole model is interpretable.

Controls. Clause-LR re-fits one *global* weight per clause (no hypernetwork); IMN-TM (rand.) replaces the learned clauses by random data-anchored conjunctions of matched length.

2.2 Training and regularisation

Stage 1 trains a weighted TM (`torchtsetlin`, 200 clauses per class, $T = 100$, $s = 5$, batch feedback with batch 32) on a thermometer encoding with up to 10 unique quantile thresholds per standardised column, keeping the validation-best epoch. Duplicate and empty clauses are removed. Stage 2 reproduces the official IMN training loop exactly: TabResNet with 2 blocks of width 128, Adam with learning rate and weight decay 0.01, 5 warm-up epochs, cosine annealing with three warm restarts, three snapshots, 500 epochs, batch 64, and an ℓ_1 penalty $\lambda\,\overline{|w(x)|}$ on the generated parameters. For the clause arms the penalty can instead shrink the generated weights toward the TM's votes, $\lambda\,\overline{|W(x) - W_{\text{TM}}|}$ (`prior`); Section 4.1 selects between the variants on validation data.

2.3 Attribution for every method

Following Eq. (2) of the paper, the impact of feature m is the change of the explained logit when x_m is set to 0 (its training mean) *with the local model held fixed*: $\phi_m(x) = z(x; w(x)) - z(x \odot \delta^m; w(x))$. For IMN this is exactly $w_m(x)x_m$ (tested); for clause local models the clauses are re-evaluated on the modified input; methods without a local model use the same removal on the full model (occlusion).

2.4 A worked example

This section follows one toy input through every step of IMN-TM, then through IMN and TM-Gated, so that each symbol in the equations above has a concrete value. The numbers are *illustrative*: they are chosen by hand to be easy to follow and are not measured. A real run differs only in size. It has tens to thousands of threshold bits, hundreds of clauses, $T = 100$ instead of $T = 10$, and weights produced by a trained TabResNet. Table 1 collects the notation and Figure 2 draws the whole computation.

The task. A binary task ($y = 1$: disease, $y = 0$: healthy) with $d = 2$ numeric features, *age* and *blood pressure* (bp). As in the official IMN pipeline, numeric columns are standardised on the training split, so a value is in units of standard deviations from the training mean. The example to explain is

$$x = (x_1, x_2) = (\underbrace{0.8}_{\text{age}},\ \underbrace{-0.3}_{\text{bp}}), \qquad y = 1,$$

a somewhat older patient with slightly low blood pressure. Binary tasks use a single output ($n_{\text{out}} = 1$), so the model produces one logit $z(x)$ and $p(y{=}1 \mid x) = \sigma(z(x))$.

Table 1: Notation, with the value each symbol takes in the worked example of Section 2.4.

Symbol	Meaning	Shape	Toy value
$x = (x_1, \ldots, x_d)$	standardised input features	d	$(0.8, -0.3)$, $d = 2$
$t_b,\ m(b)$	threshold of bit b and the feature it reads	per bit	$t_2 = 0.2$, $m(2) = 1$ (age)
$\beta_b(x) = [x_{m(b)} \geq t_b]$	Boolean (thermometer) bit	F bits	$(1, 1, 0,\ 1, 0, 0)$, $F = 6$
$\ell(x) = (\beta,\ \neg\beta)$	literals: every bit and its negation	$2F$	12 literals
$c_j(x)$	output of clause j (AND of its literals)	K	$(1, 0, 1)$, $K = 3$
W_j^{TM}	signed integer vote of clause j in the trained TM	$K \times n_{\text{out}}$	$(+6, +4, -3)$
T	TM vote margin (scales votes to logits)	scalar	10 (real runs: 100)
$W_j(x),\ b(x)$	clause weights and bias generated for x	$K \times n_{\text{out}}$, n_{out}	$(1.4,\ 0.2,\ -0.5)$, -0.3
$z(x)$	logit of the local model	n_{out}	0.6
λ	strength of the penalty on generated weights	scalar	0.1
$\phi_m(x)$	removal attribution of feature m	d	$(1.4,\ -0.2)$

Step 1: from numbers to bits. The TM reads Boolean inputs, so every feature is cut at a few thresholds (*thermometer* encoding). The thresholds are quantiles of the training data; real runs keep up to 10 per feature, and the toy keeps 3:

$$\begin{array}{lllll} \text{age } (x_1 = 0.8): & t_1 = -0.5 & t_2 = 0.2 & t_3 = 0.9 & \beta_{1..3} = (1,1,0) \\ \text{bp } (x_2 = -0.3): & t_4 = -0.6 & t_5 = 0.0 & t_6 = 0.7 & \beta_{4..6} = (1,0,0) \end{array} \qquad \beta_b(x) = [\, x_{m(b)} \geq t_b \,].$$

Here $[\cdot]$ is 1 when the condition holds and 0 otherwise, and $m(b)$ is the feature that bit b reads. Bits 1–3 have $m(b) = 1$ (age) and bits 4–6 have $m(b) = 2$ (bp). The thermometer pattern $1, 1, 0$ says "age is at least 0.2 but below 0.9". The TM may use each bit as it is or negated, which gives $2F = 12$ literals $\ell(x)$.

Step 2: the frozen clauses (stage 1). A weighted TM is trained on the bits as usual. After training, its clauses are fixed rules. Suppose that, once empty and duplicate clauses are removed, $K = 3$ remain:

	clause (in words)	$c_j(x)$	W_j^{TM}
c_1 :	$[\text{age} \geq 0.2] \ \wedge \ \neg[\text{bp} \geq 0.7]$	$1 \wedge 1 = 1$	$+6$
c_2 :	$[\text{bp} \geq 0.0]$	0	$+4$
c_3 :	$\neg[\text{age} \geq 0.9] \ \wedge \ [\text{bp} \geq -0.6]$	$1 \wedge 1 = 1$	-3

A clause outputs 1 only when *all* of its literals are 1. For a binary task, W_j^{TM} is the clause's vote for class 1 minus its vote for class 0: c_1 and c_2 vote for disease, and c_3 votes for healthy. The TM's own prediction is the vote sum scaled by the margin T:

$$z^{\text{TM}}(x) = \frac{1}{T} \sum_{j=1}^{K} W_j^{\text{TM}}\, c_j(x) = \frac{6 \cdot 1 + 4 \cdot 0 - 3 \cdot 1}{10} = 0.3, \qquad \sigma(0.3) = 0.57.$$

The weights W_j^{TM} are the *same for every patient*, which makes the TM a global model. From here on the clauses never change. For any input, $c(x) \in \{0,1\}^K$ is just a vector of K yes/no answers, and gradient descent can use it like any other fixed feature.

Step 3: the hypernetwork writes a local model (stage 2). The TabResNet hypernetwork reads the *raw* standardised input $x = (0.8, -0.3)$ and outputs $P = K + 1 = 4$ numbers: one weight per clause and a bias. For a C-class task it outputs $(K{+}1) \times C$ numbers. Suppose that for this patient it outputs

$$W(x) = (W_1(x), W_2(x), W_3(x)) = (1.4,\ 0.2,\ -0.5), \qquad b(x) = -0.3.$$

The local model is the TM with these weights in place of W^{TM}/T. This is Eq. (1) with $n_{\text{out}} = 1$:

$$z(x) = \sum_{j=1}^{K} W_j(x)\, c_j(x) + b(x) = 1.4 \cdot 1 + 0.2 \cdot 0 - 0.5 \cdot 1 - 0.3 = 0.6, \qquad \sigma(0.6) = 0.65.$$

Only *active* clauses ($c_j(x) = 1$) contribute. $W_2(x) = 0.2$ is generated, but it is multiplied by $c_2(x) = 0$ and is not part of the explanation. The explanation of this prediction reads: "*age* ≥ 0.2 *and bp* < 0.7 pushes towards disease with weight 1.4; *age* < 0.9 *and bp* ≥ -0.6 pushes towards healthy with weight 0.5; baseline -0.3." Each rule is a box in feature space.

Another patient gets other weights. For $x' = (-0.9,\ 0.8)$ the bits are $(0, 0, 0,\ 1, 1, 1)$ and the active clauses are $c(x') = (0, 1, 1)$. If the hypernetwork outputs $W(x') = (0.9, 1.1, -0.2)$ and $b(x') = 0.1$, then $z(x') = 1.1 - 0.2 + 0.1 = 1.0$. This explanation is led by the rule "*bp* ≥ 0" with weight 1.1. The global TM, by contrast, gives that rule $4/10 = 0.4$ for every patient. *Which rules count, and by how much, is decided per example; what the rules are is decided once, by the TM.*

Step 4: initialisation at the TM. Before training, the last layer of the hypernetwork is zero. Its bias is set to $W^{\text{TM}}/T = (0.6, 0.4, -0.3)$ for the clause weights and to 0 for b. The hypernetwork therefore outputs exactly these numbers for every input, and the untrained IMN-TM reproduces the TM: $z(x) = 0.6 \cdot 1 + 0.4 \cdot 0 - 0.3 \cdot 1 + 0 = 0.3 = z^{\text{TM}}(x)$. Training then only *moves* the weights away from the TM's votes where that lowers the loss. The one difference from the TM is that the TM clamps each class's vote sum to $[-T, T]$, and this initialisation does not.

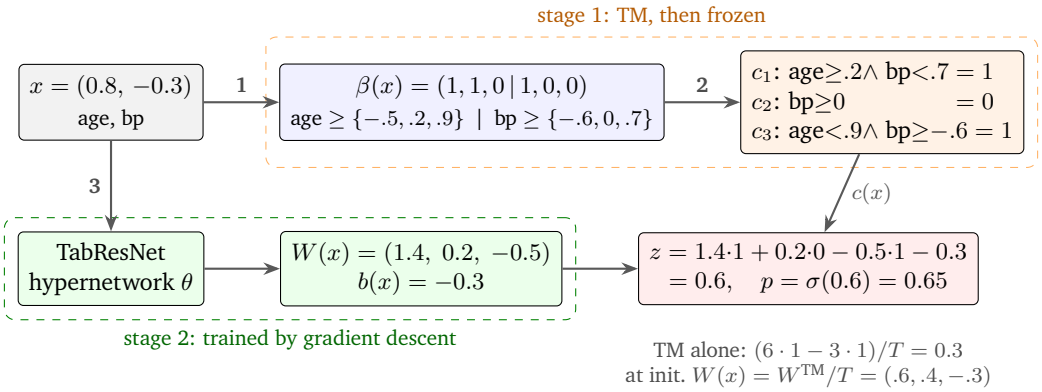


Figure 2: The worked example of Section 2.4. (1) The standardised input is cut at training-set thresholds into bits. (2) The frozen TM clauses are evaluated on the bits. (3) Separately, the hypernetwork maps the raw input to one weight per clause plus a bias. The logit is the sum of the weights of the active clauses ($c_j = 1$). Only the hypernetwork is trained in stage 2. Numbers are illustrative.

Step 5: the training loss. The hypernetwork parameters θ are trained by Adam on the cross-entropy plus a penalty on the generated parameters. For the `prior` variant the penalty pulls them towards the TM initialisation $w^{\text{prior}} = (0.6, 0.4, -0.3,\ 0)$:

$$\mathcal{L}(x, y) = \underbrace{-\log \sigma(z(x))}_{y=1:\ \log(1+e^{-0.6})=0.437} + \lambda \underbrace{\tfrac{1}{P}\textstyle\sum_{p=1}^{P} \left|w_p(x) - w_p^{\text{prior}}\right|}_{(0.8+0.2+0.2+0.3)/4=0.375} = 0.437 + 0.1 \cdot 0.375 = 0.475.$$

Here $w(x) = (W(x), b(x))$ is the full generated vector of length P, and $\lambda = 0.1$ is an example value. The gradient of $\mathcal{L}$ flows through $z(x)$ into $W(x)$ and $b(x)$, and from there into θ. It never flows into the clauses, which is why they had to be learned first. With the original `all` variant, $w^{\text{prior}} = 0$ and the penalty is $\lambda\,\overline{|w(x)|}$ as in IMN.

Step 6: feature attributions. A doctor asks how much *age* mattered. Following the paper, we set that feature to 0 (its training mean), *keep the generated weights* $W(x), b(x)$ *fixed*, re-evaluate the clauses, and measure how much the logit drops:

$$\phi_m(x) = z\big(x;\, W(x)\big) - z\big(x \odot \delta^m;\, W(x)\big),$$

where $x \odot \delta^m$ is x with feature m set to 0.

- *age* $\to 0$: $x = (0, -0.3)$ has bits $(1, 0, 0,\ 1, 0, 0)$. Now $c_1 = 0$ (age < 0.2) while c_3 stays on, so $z = -0.5 - 0.3 = -0.8$ and $\phi_{\text{age}} = 0.6 - (-0.8) = 1.4$.
- *bp* $\to 0$: $x = (0.8, 0)$ has bits $(1, 1, 0,\ 1, 1, 0)$. Now c_2 switches *on*, so $z = 1.4 + 0.2 - 0.5 - 0.3 = 0.8$ and $\phi_{\text{bp}} = 0.6 - 0.8 = -0.2$.

Age therefore pushes this patient towards disease, and bp pushes slightly towards healthy, because raising bp to average would fire the disease rule c_2. A feature's attribution sums the effect of every rule that it switches on or off. This is why a clause explanation stays correct under small changes of x (local fidelity, Section 4.3). A rule only changes its contribution when x crosses one of its thresholds.

The same patient under IMN and TM-Gated. The original IMN hypernetwork outputs a linear model, for example $w(x) = (1.2,\ 0.5)$ and $w_0(x) = -0.1$, so

$$z(x) = w_1(x)x_1 + w_2(x)x_2 + w_0(x) = 1.2{\cdot}0.8 + 0.5{\cdot}(-0.3) - 0.1 = 0.71, \qquad \phi(x) = (w_1 x_1,\ w_2 x_2) = (0.96,\ -0.15).$$

The explanation is two numbers instead of two rules, and it assumes the logit changes linearly in age and bp around x. In TM-Gated there is no TabResNet: the coefficients of the linear model are themselves a TM-style vote, $w(x) = w_0 + \sum_j c_j(x) V_j$, where row V_j is what clause j adds to the coefficients (age, bp, bias) when it fires. With $w_0 = (0.3,\ 0.1,\ 0)$, $V_1 = (0.8,\ 0,\ 0.1)$ and $V_3 = (0,\ -0.4,\ 0)$, the active clauses c_1, c_3 give $w(x) = (1.1,\ -0.3,\ 0.1)$ and $z(x) = 1.1 \cdot 0.8 - 0.3 \cdot (-0.3) + 0.1 = 1.07$. Every coefficient can be traced to a global value plus the rules that fired. IMN-Hybrid simply adds the IMN linear term and the IMN-TM clause term before the shared bias.

3 Experimental protocol

Data. Tier 1 is 12 of the paper's 35 AutoML-benchmark datasets (5–145 features, 690–48,842 rows, 3 multi-class); Tier 2 the other 23. Preprocessing follows the official code (standardisation, target encoding for binary

and ordinal encoding for multi-class tasks). Every method sees the same stratified 60/20/20 train/validation/test split per seed; the TM needs a validation split, and giving it to all methods keeps the comparison fair, but it also means our numbers are not directly comparable to the paper's 80/20 protocol.

Methods. LR, DT, RF (500 trees), GBDT (sklearn HistGradientBoosting; CatBoost was unavailable), TabResNet, IMN, TM, and the TM-based arms of Section 2.

Statistics. The unit of analysis is the dataset-level mean test AUROC over seeds. We report average ranks with a Friedman test and Nemenyi critical difference (as in the paper) and preregistered two-sided Wilcoxon signed-rank tests with Holm correction, plus a paired bootstrap CI of the mean difference. Non-inferiority of IMN-TM to IMN was preregistered as a 95% CI above -0.01.

Rules. The library was not modified; every number is generated from the run records; all selection uses validation data only. Deviations from the paper are logged in `DECISIONS.md`.

4 Results

4.1 Phase 1: regulariser of the clause arms

Table 2: Mean validation AUROC over 6 datasets $\times$ 3 seeds (324 runs) for the nine penalty settings. The selected setting is `prior`, $\lambda = 1$.

ℓ_1 mode, λ	IMN-TM	IMN-Hybrid
`prior`, 1	0.8508	0.8510
`all`, 1	0.8485	0.8501
`all`, 0.1	0.8461	0.8454
`active`, 1	0.8456	0.8464
`prior`, 0.1	0.8456	0.8454
`prior`, 0.01	0.8440	0.8442
`active`, 0.1	0.8438	0.8447
`all`, 0.01	0.8433	0.8454
`active`, 0.01	0.8425	0.8444

Shrinking toward the TM's votes with the largest weight tried wins for both clause arms (Table 2), but all nine settings lie within a few thousandths of AUROC, and the winner sits at the edge of the grid; the tuned stage therefore searches $\lambda \in [10^{-5}, 10]$.

4.2 Main benchmark, default hyper-parameters

Table 3 gives the per-dataset results and Table 4 the preregistered tests.

- **Q1 (vs. the TM).** IMN-TM improves on its own source TM on 11 of 12 datasets (mean +0.0074, CI [+0.0038, +0.0114], Holm $p = 0.024$), the only significant comparison.
- **Q2 (vs. IMN).** Mean difference +0.0006 with 6/6 wins/losses; the CI [-0.0140, +0.0160] is too wide to establish the preregistered non-inferiority margin, so the question is open in both directions.
- **Q3–Q6.** The hybrid behaves like IMN-TM (+0.0005); TM-Gated is level with LR, the TM and IMN; the per-example generator adds +0.0024 over a global re-fit of the clause weights; random clauses reach the same AUROC as learned ones (-0.0009).
- **Reproduction.** IMN and TabResNet are statistically indistinguishable (+0.0026), as in the paper; the largest gaps to the paper's own numbers are phoneme (-0.068), jungle_chess (-0.061).

Average ranks are in Figure 5: RF (3.75) and GBDT (3.92) lead; all mesomorphic variants lie between 5.00 and 6.67 (Friedman $p = 5.6 \times 10^{-7}$, critical difference 4.81).

The gain is predictable. The average hides a strong structure (Figure 3): the per-dataset gain of IMN-TM over IMN correlates with the gap between the best tree ensemble and the best linear model at $r = 0.92$. Where axis-aligned thresholds and interactions matter (phoneme, jungle-chess, kc1), conjunctive local models beat hyperplanes; where the problem is smooth or nearly linear (blood-transfusion, vehicle, credit-g), they lose. The hybrid was meant to cover both cases but does not; Section 4.7 tests whether the strong prior penalty suppresses its linear term.

Table 3: Test AUROC (mean$_{\pm sd}$ over 5 seeds), default hyper-parameters. Top: baselines; bottom: TM family, with IMN repeated for reference. Bold: best within each half. Last column: the paper's Table 8 IMN value (its own split; reference only).

Dataset	LR	DT	RF	GBDT	TabResNet	IMN
kr-vs-kp	0.990$_{\pm 0.003}$	0.985$_{\pm 0.003}$	0.998$_{\pm 0.001}$	**1.000$_{\pm 0.000}$**	0.997$_{\pm 0.002}$	0.997$_{\pm 0.002}$
credit-g	0.777$_{\pm 0.053}$	0.634$_{\pm 0.046}$	0.798$_{\pm 0.047}$	0.787$_{\pm 0.045}$	0.798$_{\pm 0.047}$	**0.799$_{\pm 0.044}$**
vehicle	0.937$_{\pm 0.009}$	0.799$_{\pm 0.016}$	0.929$_{\pm 0.004}$	0.925$_{\pm 0.011}$	0.957$_{\pm 0.007}$	**0.957$_{\pm 0.008}$**
kc1	0.813$_{\pm 0.012}$	0.638$_{\pm 0.044}$	**0.827$_{\pm 0.007}$**	0.797$_{\pm 0.014}$	0.810$_{\pm 0.018}$	0.812$_{\pm 0.005}$
blood-transfusion	0.739$_{\pm 0.037}$	0.610$_{\pm 0.042}$	0.665$_{\pm 0.052}$	0.680$_{\pm 0.072}$	0.738$_{\pm 0.052}$	**0.742$_{\pm 0.047}$**
phoneme	0.810$_{\pm 0.009}$	0.822$_{\pm 0.017}$	**0.957$_{\pm 0.006}$**	0.950$_{\pm 0.006}$	0.867$_{\pm 0.007}$	0.882$_{\pm 0.005}$
adult	0.902$_{\pm 0.003}$	0.753$_{\pm 0.006}$	0.917$_{\pm 0.002}$	**0.927$_{\pm 0.001}$**	0.905$_{\pm 0.003}$	0.904$_{\pm 0.002}$
Australian	0.922$_{\pm 0.019}$	0.824$_{\pm 0.039}$	**0.941$_{\pm 0.010}$**	0.932$_{\pm 0.020}$	0.928$_{\pm 0.020}$	0.923$_{\pm 0.022}$
segment	0.993$_{\pm 0.002}$	0.981$_{\pm 0.003}$	0.998$_{\pm 0.001}$	**0.999$_{\pm 0.000}$**	0.995$_{\pm 0.001}$	0.995$_{\pm 0.001}$
jungle_chess	0.805$_{\pm 0.003}$	0.798$_{\pm 0.006}$	0.943$_{\pm 0.002}$	**0.974$_{\pm 0.001}$**	0.903$_{\pm 0.008}$	0.917$_{\pm 0.003}$
jasmine	0.853$_{\pm 0.009}$	0.749$_{\pm 0.015}$	**0.879$_{\pm 0.006}$**	0.864$_{\pm 0.003}$	0.858$_{\pm 0.011}$	0.859$_{\pm 0.011}$
sylvine	0.966$_{\pm 0.003}$	0.913$_{\pm 0.011}$	0.981$_{\pm 0.002}$	**0.984$_{\pm 0.002}$**	0.976$_{\pm 0.003}$	0.977$_{\pm 0.002}$

Dataset	IMN	TM	Clause-LR	IMN-TM	IMN-TM (rand.)	IMN-Hybrid	TM-Gated	Paper IMN
kr-vs-kp	0.997$_{\pm 0.002}$	0.996$_{\pm 0.004}$	0.997$_{\pm 0.002}$	0.998$_{\pm 0.002}$	0.998$_{\pm 0.002}$	0.998$_{\pm 0.002}$	**0.998$_{\pm 0.002}$**	0.999
credit-g	**0.799$_{\pm 0.044}$**	0.779$_{\pm 0.043}$	0.768$_{\pm 0.040}$	0.777$_{\pm 0.036}$	0.756$_{\pm 0.038}$	0.774$_{\pm 0.035}$	0.754$_{\pm 0.034}$	0.751
vehicle	**0.957$_{\pm 0.008}$**	0.913$_{\pm 0.020}$	0.919$_{\pm 0.020}$	0.922$_{\pm 0.019}$	0.939$_{\pm 0.011}$	0.926$_{\pm 0.017}$	0.918$_{\pm 0.016}$	0.957
kc1	0.812$_{\pm 0.005}$	0.813$_{\pm 0.016}$	0.826$_{\pm 0.008}$	0.831$_{\pm 0.012}$	0.829$_{\pm 0.007}$	**0.834$_{\pm 0.013}$**	0.785$_{\pm 0.012}$	0.805
blood-transfusion	**0.742$_{\pm 0.047}$**	0.696$_{\pm 0.054}$	0.694$_{\pm 0.069}$	0.698$_{\pm 0.066}$	0.709$_{\pm 0.063}$	0.700$_{\pm 0.064}$	0.681$_{\pm 0.083}$	0.742
phoneme	0.882$_{\pm 0.005}$	0.918$_{\pm 0.006}$	0.927$_{\pm 0.007}$	0.926$_{\pm 0.007}$	0.917$_{\pm 0.005}$	0.926$_{\pm 0.007}$	**0.935$_{\pm 0.008}$**	0.950
adult	0.904$_{\pm 0.002}$	0.876$_{\pm 0.003}$	0.887$_{\pm 0.002}$	0.899$_{\pm 0.003}$	0.904$_{\pm 0.002}$	0.900$_{\pm 0.003}$	**0.906$_{\pm 0.002}$**	0.915
Australian	0.923$_{\pm 0.022}$	0.922$_{\pm 0.019}$	0.934$_{\pm 0.017}$	0.932$_{\pm 0.019}$	**0.935$_{\pm 0.022}$**	0.933$_{\pm 0.021}$	0.926$_{\pm 0.022}$	0.908
segment	0.995$_{\pm 0.001}$	0.997$_{\pm 0.001}$	0.998$_{\pm 0.000}$	0.998$_{\pm 0.000}$	0.998$_{\pm 0.001}$	**0.998$_{\pm 0.000}$**	0.998$_{\pm 0.001}$	0.994
jungle_chess	0.917$_{\pm 0.003}$	0.958$_{\pm 0.003}$	0.965$_{\pm 0.002}$	0.965$_{\pm 0.003}$	0.956$_{\pm 0.003}$	0.965$_{\pm 0.002}$	**0.972$_{\pm 0.002}$**	0.978
jasmine	0.859$_{\pm 0.011}$	0.846$_{\pm 0.006}$	0.851$_{\pm 0.006}$	0.852$_{\pm 0.012}$	**0.864$_{\pm 0.015}$**	0.850$_{\pm 0.010}$	0.838$_{\pm 0.008}$	0.865
sylvine	0.977$_{\pm 0.002}$	0.968$_{\pm 0.002}$	0.974$_{\pm 0.003}$	0.972$_{\pm 0.002}$	0.976$_{\pm 0.003}$	0.973$_{\pm 0.002}$	**0.978$_{\pm 0.002}$**	0.981

4.3 Interpretability

Clause local models describe the model around an example far better than IMN's hyperplanes (Table 5, Figure 4): the local model of an example classifies its 10 nearest neighbours per class with accuracy 0.72 for IMN-TM vs. 0.55 for IMN, and the frozen local model agrees with the full model's response to feature removal (self-consistency 0.96 vs. 0.65). Learned clauses matter here even though they do not matter for AUROC: random clauses drop to 0.54 and 0.69. The cost is length: 51 weighted rules are needed to cover 90% of an IMN-TM explanation vs. 15 features for IMN. TM-Gated gives short linear explanations (17 terms) but the lowest fidelity (0.45): its local hyperplane changes whenever a rule switches. Figure 6 shows both kinds of explanation for one real decision.

4.4 Tuned hyper-parameters

Every method received 20 random-search trials per dataset (§3). The winner of each method and dataset was chosen on the seed-0 validation AUROC only and then re-run on seeds 0–4. Tables 6 and 7 repeat the main-stage analysis on these 720 runs.

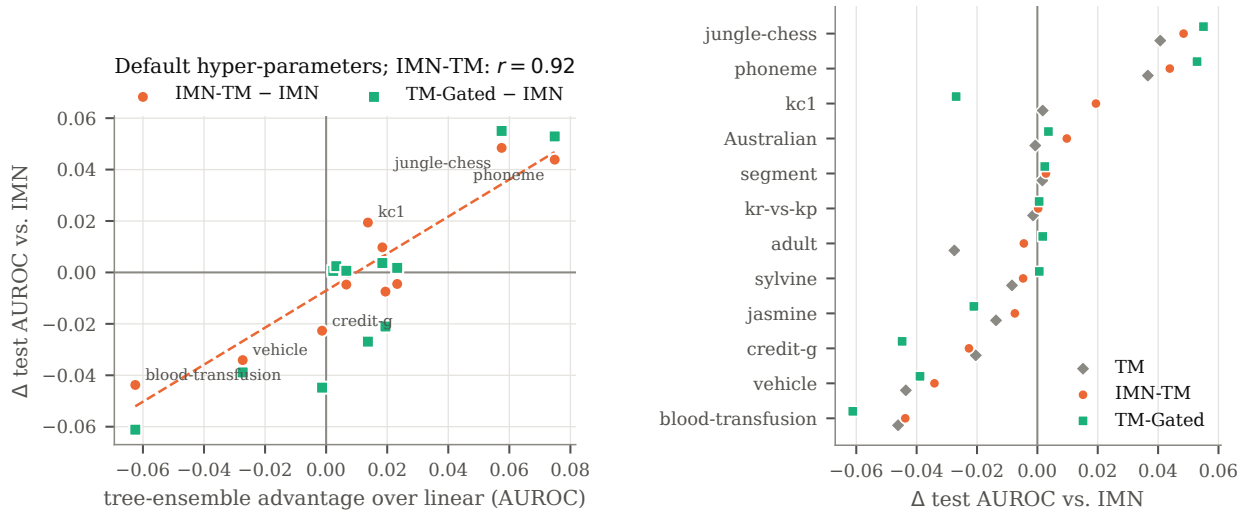
Tuning helps the clause models most. IMN-TM improves on its default configuration on 11 of 12 datasets (+0.0095 mean AUROC, to 0.9071). IMN gains +0.0043 on 8 datasets and TabResNet gains +0.0131. Random forests lose -0.0064, which shows that selecting on one validation split is noisy. After tuning, IMN-TM shares the best average rank with GBDT (both 3.75; IMN-Hybrid 3.92, TabResNet 4.50, IMN 7.25; Figure 5). The critical difference is 4.81, however, so the leading group cannot be separated statistically.

Preregistered questions.

- **Q1** is stronger than with defaults: IMN-TM beats its source TM on 12/12 datasets (+0.0094, Holm $p = 0.005$).
- **Q5** becomes significant. The per-example generator beats a single global re-fit of the same clause weights by +0.0036 (10/2, Holm $p = 0.031$).
- **Q2** moves in favour of IMN-TM (+0.0058, 9/3) but remains unresolved: the confidence interval includes zero. One dataset dominates the mean: on blood-transfusion (+0.037) the tuned IMN lost blood-transfusion

Table 4: Preregistered comparisons, default hyper-parameters (12 datasets). *: Holm-corrected $p < 0.05$.

Q	Comparison A vs. B	mean $A-B$	95% bootstrap CI	W/L	p	p_{Holm}
Q1	IMN-TM vs. TM	+0.0074	[+0.0038, +0.0114]	11/1	0.002	0.024*
Q2	IMN-TM vs. IMN	+0.0006	[-0.0140, +0.0160]	6/6	1.000	1.000
Q3	IMN-Hybrid vs. IMN	+0.0011	[-0.0134, +0.0163]	6/6	1.000	1.000
Q3	IMN-Hybrid vs. IMN-TM	+0.0005	[-0.0004, +0.0014]	9/3	0.204	1.000
Q4	TM-Gated vs. LR	+0.0151	[-0.0158, +0.0538]	7/5	0.970	1.000
Q4	TM-Gated vs. TM	+0.0004	[-0.0090, +0.0094]	8/4	0.733	1.000
Q4	TM-Gated vs. IMN	-0.0063	[-0.0250, +0.0134]	7/5	0.850	1.000
Q5	IMN-TM vs. Clause-LR	+0.0024	[+0.0002, +0.0050]	9/3	0.110	0.879
Q6	IMN-TM vs. IMN-TM (rand.)	-0.0009	[-0.0061, +0.0049]	5/7	0.569	1.000
R	IMN vs. TabResNet	+0.0026	[-0.0004, +0.0061]	9/3	0.092	0.831



(a) Gain over IMN vs. the tree-ensemble advantage over linear models (one point per dataset).

(b) Per-dataset ΔAUROC vs. IMN, sorted by the IMN-TM gain.

Figure 3: Where the TM helps. Default hyper-parameters, mean over 5 seeds.

(-0.073) against its own default, a selection failure on a dataset of 748 rows. Without it the mean difference is +0.0029.

- **Q6** is unchanged. Random clauses remain as accurate as learned ones (+0.0007).
- **R**: tuned IMN falls behind tuned TabResNet (-0.0061, 1/11, Holm $p = 0.215$). The paper reports them level.

The default-HP pattern does not survive tuning. With tuning, IMN gains most on exactly the datasets where clauses had helped: jungle_chess (+0.063), phoneme (+0.063). The correlation between the IMN-TM gain and the tree-ensemble advantage falls from $r = 0.92$ to $r = -0.11$ (Figure 7). The default-HP result should therefore be read as follows. The clause basis gives the hypernetwork a good inductive bias for threshold-shaped problems without any search. A tuned hyperplane model closes most of that gap, at a much higher search cost.

Explanations after tuning. Local fidelity is unchanged (IMN-TM 0.72 vs. IMN 0.57) and self-consistency stays high (0.90). The tuned clause models choose weaker L1 penalties, though, and their explanations grow from 51 to 126 rules for 90% of the evidence. An accuracy-only search therefore trades away explanation length; a search that also scores explanation size would need an explicit constraint.

4.5 Tier 2: twenty larger datasets

Tier 2 covers the remaining AutoML-benchmark datasets. They have up to 324k training rows, up to 4,296 features and up to 100 classes. Three are excluded (D25): guillermo and robert fail OpenML's checksum, and albert cannot be downloaded. The remaining 20 datasets are run with 3 seeds and default hyper-parameters (720 runs). All runs completed; 0 were skipped for size.

Two preregistered caps keep the cost bounded (D24):

Table 5: Explanation size and validity (500 test examples per run, mean over datasets and seeds). "terms": number of terms covering 90% of the absolute contribution; self-consist.: cosine between the frozen-local-model attribution and the removal effect on the full model; fid.: accuracy of an example's local model on its k nearest test neighbours per class.

Method	lin. terms	clause terms	active clauses	literals/clause	self-consist.	fid. k=10	fid. k=50
IMN	14.9	–	–	–	0.649	0.553	0.581
Clause-LR	–	51.8	82.6	6.7	1.000	0.736	0.777
IMN-TM	–	51.0	82.6	6.7	0.965	0.724	0.764
IMN-TM (rand.)	–	40.4	79.5	4.4	0.686	0.538	0.554
IMN-Hybrid	9.3	50.6	82.6	6.7	0.972	0.733	0.773
TM-Gated	16.8	–	–	–	0.543	0.453	0.476

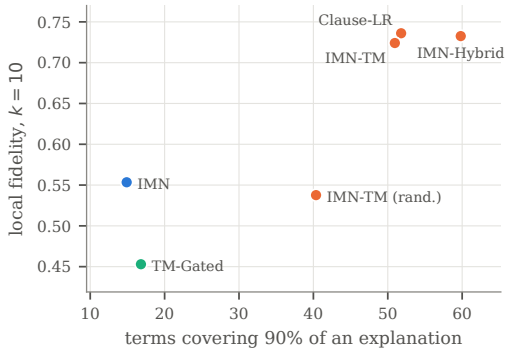


Figure 4: Explanation length vs. local fidelity.

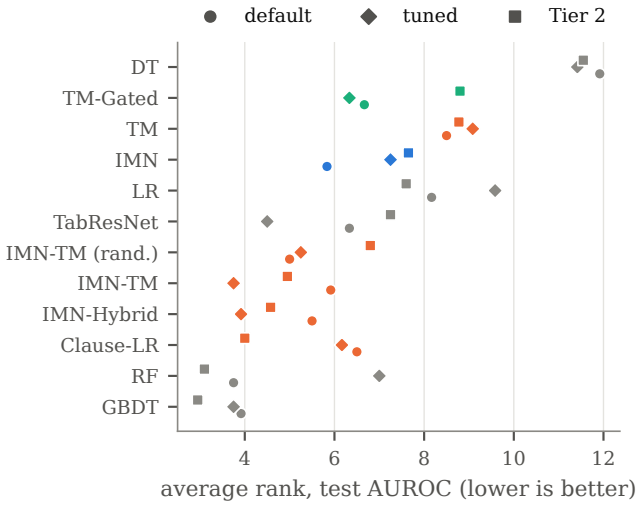


Figure 5: Average ranks (test AUROC) per protocol.

- neural training stops after 50,000 steps;
- the source TM trains for at most 20 epochs.

The step cap binds on 15 of the 20 datasets. Tables 8 and 9 report the results.

Headline. On Tier 2 every clause-based question that was open on Tier 1 comes out significant:

- IMN-TM beats its TM on 19/20 datasets (+0.0372, Holm $p = 9.5 \times 10^{-5}$);
- it beats IMN on 15/20 (+0.0183, Holm $p = 0.030$);
- learned clauses beat random ones (+0.0133, 16/4, Holm $p = 0.005$).

Most of the gain comes from an under-trained IMN. Our IMN falls well short of the paper's own numbers (Table 8, last column) on 6 datasets: higgs (-0.118), shuttle (-0.087), MiniBooNE (-0.077), KDDCup09_appetency (-0.057), volkert (-0.052), jannis (-0.040). All 6 of them are datasets where the step cap stopped training early. They are also the datasets with the largest IMN-TM gains. Across datasets, the IMN-TM−IMN difference correlates with IMN's shortfall against the paper at $r = -0.88$.

The likely mechanism is the initialisation. IMN-TM starts from a trained TM's votes, so a truncated schedule costs it little. IMN starts from scratch and loses most of its cosine schedule.

On the other 14 datasets the advantage is +0.0027 (9/5, Wilcoxon $p = 0.194$). This is the same small, non-significant edge as on Tier 1 (§4.2, §4.4). The Tier-2 result is therefore evidence that IMN-TM is the better choice *under a fixed training budget*. It is not evidence that the clause basis is a better local model.

The learned-vs-random gap (Q6) has the same caveat: random clauses give a weaker starting point.

Other observations.

- **Clause-LR.** A single global re-fit of the clause weights ranks third (4.00). It is ahead of both mesomorphic clause models (IMN-Hybrid 4.58, IMN-TM 4.95) and far ahead of IMN (7.65). On large datasets, a fixed linear model over trained clauses is a strong and cheap interpretable baseline.
- **Leaders.** GBDT (2.95) and RF (3.10) lead (Friedman $p = 9.4 \times 10^{-20}$, CD 3.73).

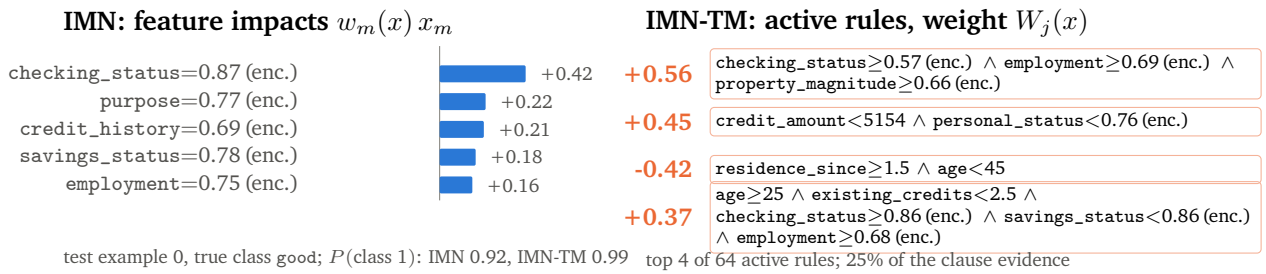
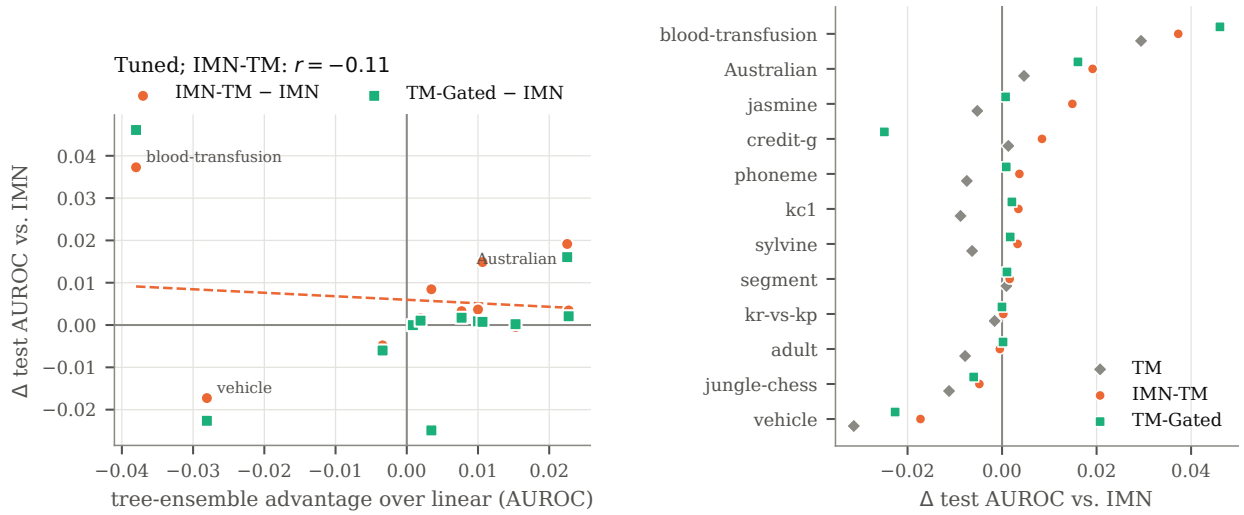


Figure 6: One credit-g decision, two explanations (seed-0 models of Table 3). Left: IMN's feature impacts. Right: the four strongest of the active rules of IMN-TM with their per-example weights; thermometer literals of one column are merged into an interval. "(enc.)": categorical feature in its target-encoded value.



(a) Gain over IMN vs. the tree-ensemble advantage, tuned.

(b) Per-dataset ΔAUROC vs. IMN, tuned.

Figure 7: Where the TM helps, after tuning. Compare with Figure 3.

- **Explanation length.** On these wider inputs, an IMN explanation needs 257 features for 90% of its evidence, and an IMN-TM explanation 85 clauses.
- **Local fidelity.** IMN-TM 0.67 vs. IMN 0.64.

4.6 Synthetic attribution benchmark

On real data the true reason for a decision is unknown. Phase 4 therefore uses three synthetic families whose per-feature contributions are known by construction. Each has $d = 8$ features and $n = 10{,}000$ examples:

- Gaussian linear;
- non-linear additive;
- piecewise constant, i.e. axis-aligned thresholds.

Each family is generated at feature correlations $\rho \in \{0, 0.25, 0.5, 0.75\}$ with five seeds, giving 60 tasks and 540 runs. Every method is explained with the same removal attribution (§2) on 500 test rows. The headline metric is the Pearson correlation between these attributions and the true contributions ("GT corr.", Table 10, Figure 8).

IMN-TM explanations are closer to the truth than IMN's. Paired over the 60 tasks, the attributions of IMN-TM correlate better with the true contributions than those of IMN on 41 of 60 tasks (mean +0.050, Wilcoxon $p = 2.4 \times 10^{-4}$).

- **Piecewise constant.** The difference is clearest here: 0.616 for IMN-TM vs. 0.570 for IMN. In this family IMN is also less accurate (AUROC 0.848 vs. 0.944).
- **Gaussian linear.** Surprisingly, the difference also holds on the linear family (0.919 vs. 0.843). As ρ rises from 0 to 0.75, IMN's correlation drops by -0.086 and IMN-TM's by only -0.003. With correlated inputs, a per-example hyperplane can shift weight between correlated features without changing its output. The TM-initialised clause weights are anchored to a global rule set and drift less.

Table 6: Test AUROC (mean$_{\pm sd}$ over 5 seeds) with tuned hyper-parameters. Layout as in Table 3.

Dataset	LR	DT	RF	GBDT	TabResNet	IMN
kr-vs-kp	0.994$_{\pm 0.002}$	0.995$_{\pm 0.002}$	0.995$_{\pm 0.002}$	**0.999**$_{\pm 0.000}$	0.999$_{\pm 0.001}$	0.999$_{\pm 0.001}$
credit-g	0.788$_{\pm 0.052}$	0.716$_{\pm 0.052}$	0.794$_{\pm 0.052}$	0.789$_{\pm 0.057}$	**0.796**$_{\pm 0.048}$	0.791$_{\pm 0.045}$
vehicle	0.944$_{\pm 0.009}$	0.854$_{\pm 0.018}$	0.925$_{\pm 0.005}$	0.924$_{\pm 0.011}$	**0.958**$_{\pm 0.007}$	0.953$_{\pm 0.011}$
kc1	0.811$_{\pm 0.013}$	0.749$_{\pm 0.037}$	**0.835**$_{\pm 0.007}$	0.820$_{\pm 0.008}$	0.814$_{\pm 0.012}$	0.813$_{\pm 0.010}$
blood-transfusion	**0.737**$_{\pm 0.032}$	0.688$_{\pm 0.048}$	0.699$_{\pm 0.037}$	0.694$_{\pm 0.038}$	0.722$_{\pm 0.054}$	0.669$_{\pm 0.058}$
phoneme	0.810$_{\pm 0.009}$	0.869$_{\pm 0.011}$	0.919$_{\pm 0.007}$	**0.954**$_{\pm 0.005}$	0.936$_{\pm 0.005}$	0.944$_{\pm 0.005}$
adult	0.902$_{\pm 0.003}$	0.900$_{\pm 0.003}$	0.910$_{\pm 0.003}$	**0.928**$_{\pm 0.001}$	0.914$_{\pm 0.002}$	0.912$_{\pm 0.002}$
Australian	0.921$_{\pm 0.003}$	0.921$_{\pm 0.016}$	0.937$_{\pm 0.012}$	**0.943**$_{\pm 0.016}$	0.927$_{\pm 0.020}$	0.920$_{\pm 0.023}$
segment	0.995$_{\pm 0.001}$	0.989$_{\pm 0.003}$	0.998$_{\pm 0.001}$	**0.999**$_{\pm 0.000}$	0.998$_{\pm 0.000}$	0.997$_{\pm 0.001}$
jungle_chess	0.803$_{\pm 0.003}$	0.856$_{\pm 0.005}$	0.896$_{\pm 0.003}$	0.977$_{\pm 0.001}$	**0.981**$_{\pm 0.001}$	0.980$_{\pm 0.001}$
jasmine	0.854$_{\pm 0.008}$	0.838$_{\pm 0.007}$	**0.871**$_{\pm 0.005}$	0.866$_{\pm 0.009}$	0.865$_{\pm 0.009}$	0.861$_{\pm 0.012}$
sylvine	0.966$_{\pm 0.003}$	0.957$_{\pm 0.009}$	0.975$_{\pm 0.001}$	**0.985**$_{\pm 0.002}$	0.980$_{\pm 0.003}$	0.977$_{\pm 0.002}$

Dataset	IMN	TM	Clause-LR	IMN-TM	IMN-TM (rand.)	IMN-Hybrid	TM-Gated	Paper IMN
kr-vs-kp	0.999$_{\pm 0.001}$	0.997$_{\pm 0.002}$	0.999$_{\pm 0.001}$	**0.999**$_{\pm 0.001}$	**0.999**$_{\pm 0.001}$	0.999$_{\pm 0.001}$	0.998$_{\pm 0.001}$	0.999
credit-g	0.791$_{\pm 0.045}$	0.792$_{\pm 0.051}$	0.793$_{\pm 0.051}$	0.799$_{\pm 0.049}$	0.792$_{\pm 0.061}$	**0.799**$_{\pm 0.049}$	0.766$_{\pm 0.030}$	0.751
vehicle	**0.953**$_{\pm 0.011}$	0.922$_{\pm 0.009}$	0.927$_{\pm 0.008}$	0.936$_{\pm 0.011}$	0.937$_{\pm 0.005}$	0.937$_{\pm 0.011}$	0.931$_{\pm 0.010}$	0.957
kc1	0.813$_{\pm 0.010}$	0.804$_{\pm 0.013}$	0.816$_{\pm 0.007}$	0.816$_{\pm 0.008}$	0.817$_{\pm 0.010}$	**0.818**$_{\pm 0.009}$	0.815$_{\pm 0.015}$	0.805
blood-transfusion	0.669$_{\pm 0.058}$	0.698$_{\pm 0.059}$	0.705$_{\pm 0.065}$	0.706$_{\pm 0.065}$	**0.725**$_{\pm 0.057}$	0.708$_{\pm 0.066}$	0.715$_{\pm 0.051}$	0.742
phoneme	0.944$_{\pm 0.005}$	0.937$_{\pm 0.005}$	0.943$_{\pm 0.004}$	0.948$_{\pm 0.004}$	0.945$_{\pm 0.004}$	**0.948**$_{\pm 0.004}$	0.945$_{\pm 0.004}$	0.950
adult	0.912$_{\pm 0.002}$	0.905$_{\pm 0.003}$	0.910$_{\pm 0.003}$	0.912$_{\pm 0.002}$	0.910$_{\pm 0.002}$	0.912$_{\pm 0.002}$	**0.913**$_{\pm 0.002}$	0.915
Australian	0.920$_{\pm 0.023}$	0.924$_{\pm 0.012}$	0.938$_{\pm 0.010}$	**0.939**$_{\pm 0.006}$	0.938$_{\pm 0.013}$	0.939$_{\pm 0.005}$	0.936$_{\pm 0.011}$	0.908
segment	0.997$_{\pm 0.001}$	0.998$_{\pm 0.001}$	**0.999**$_{\pm 0.000}$	0.998$_{\pm 0.000}$	0.997$_{\pm 0.001}$	0.996$_{\pm 0.002}$	0.998$_{\pm 0.002}$	0.994
jungle_chess	**0.980**$_{\pm 0.001}$	0.969$_{\pm 0.001}$	0.974$_{\pm 0.001}$	0.975$_{\pm 0.002}$	0.970$_{\pm 0.001}$	0.975$_{\pm 0.002}$	0.974$_{\pm 0.002}$	0.978
jasmine	0.861$_{\pm 0.012}$	0.856$_{\pm 0.012}$	0.863$_{\pm 0.008}$	**0.876**$_{\pm 0.009}$	0.869$_{\pm 0.013}$	0.876$_{\pm 0.009}$	0.862$_{\pm 0.009}$	0.865
sylvine	0.977$_{\pm 0.002}$	0.971$_{\pm 0.005}$	0.976$_{\pm 0.003}$	0.981$_{\pm 0.002}$	0.978$_{\pm 0.003}$	**0.981**$_{\pm 0.002}$	0.979$_{\pm 0.002}$	0.981

Baselines. The best global models remain ahead.
- Plain logistic regression is nearly perfect on its own family (0.975).
- GBDT's removal attributions beat IMN-TM on 50 of the 60 tasks (-0.026, $p = 1.5 \times 10^{-6}$).

The mesomorphic models are thus not more faithful than a removal-attributed tree ensemble. Their advantage is that the explanation is the model's own local function, not a post-hoc probe.

TM-Gated explanations are unreliable. TM-Gated is the least faithful method in every family (0.229 on piecewise data; IMN-TM is ahead on 59/60 tasks). Its local hyperplane is re-composed from rule-gated pieces, and the removal of one feature switches rules on or off. The same failure appeared as low local fidelity on real data (§4.3).

4.7 Ablations

Each variant of Table 11 is run on the twelve Tier-1 datasets with seeds 0–2. It is compared with the main-stage record of its reference arm on the same seeds, using a paired Wilcoxon test over datasets without multiplicity correction. No variant changes test AUROC significantly. Read together, the rows answer four design questions.

Basis size. The source TM gains from more clauses (800 per class: +0.0036, 9/3 wins/losses, $p = 0.077$). IMN-TM does not (-0.0012, $p = 0.622$), and 50 clauses per class are also enough (+0.0004, $p = 0.850$). Re-weighting the clauses per example therefore recovers what a larger TM gets by adding clauses, and the default of 200 is not a binding choice.

Initialisation. Starting the generated weights at zero instead of at the TM's own votes leaves accuracy unchanged (+0.0002, $p = 0.970$). It does lower local fidelity from 0.73 to 0.65. The initialisation is therefore an interpretability device: it keeps the per-example weights close to a global rule set.

Generator input. Feeding the clause outputs to the hypernetwork as well, $[x, c(x)]$, is the worst variant (-0.0116, 5/7, $p = 0.064$). The generator then sees the same binary evidence it is weighting, which appears to encourage over-fitting. The raw input x alone is the better choice.

Table 7: Preregistered comparisons, tuned hyper-parameters (12 datasets). *: Holm-corrected $p < 0.05$.

Q	Comparison A vs. B	mean $A-B$	95% bootstrap CI	W/L	p	p_{Holm}
Q1	IMN-TM vs. TM	+0.0094	[+0.0065, +0.0125]	12/0	4.9×10^{-4}	0.005*
Q2	IMN-TM vs. IMN	+0.0058	[-0.0010, +0.0136]	9/3	0.129	0.776
Q3	IMN-Hybrid vs. IMN	+0.0059	[-0.0010, +0.0139]	8/4	0.176	0.776
Q3	IMN-Hybrid vs. IMN-TM	+0.0001	[-0.0004, +0.0006]	7/5	0.380	0.904
Q4	TM-Gated vs. LR	+0.0254	[-0.0031, +0.0629]	9/3	0.301	0.904
Q4	TM-Gated vs. TM	+0.0049	[-0.0016, +0.0096]	11/1	0.034	0.239
Q4	TM-Gated vs. IMN	+0.0013	[-0.0075, +0.0116]	8/4	0.519	0.904
Q5	IMN-TM vs. Clause-LR	+0.0036	[+0.0015, +0.0060]	10/2	0.003	0.031*
Q6	IMN-TM vs. IMN-TM (rand.)	+0.0007	[-0.0035, +0.0037]	8/4	0.151	0.776
R	IMN vs. TabResNet	-0.0061	[-0.0155, -0.0001]	1/11	0.027	0.215

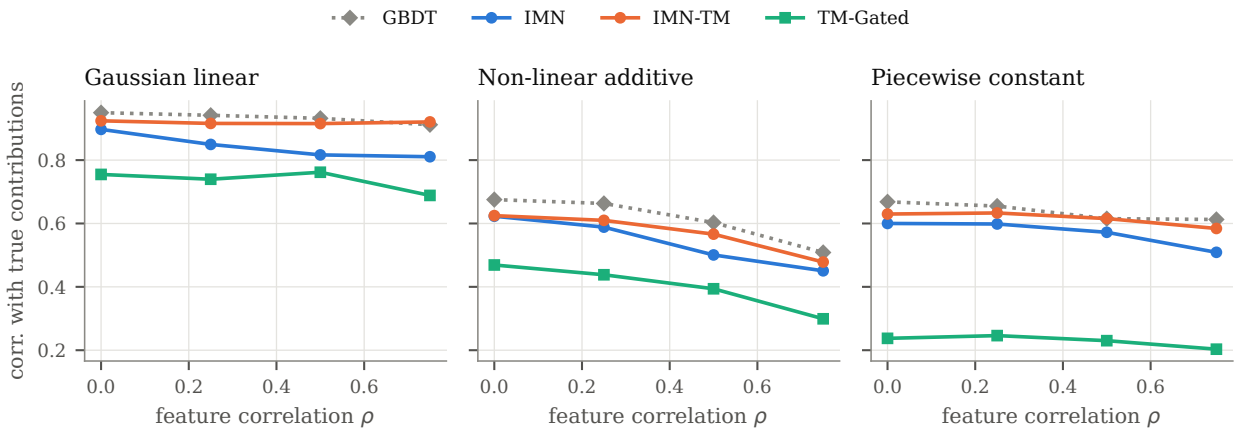


Figure 8: Attribution quality vs. feature correlation. Mean over five seeds.

Which clauses. Random conjunctions matched to the trained clauses' firing rates give nearly the same accuracy (-0.0017, $p = 0.850$), confirming the main-stage random-clause control. Their local fidelity, however, drops from 0.73 to 0.56. The trained TM is not needed for accuracy, but it is what makes the explanations generalise to the neighbourhood of an example.

Hybrid split. Giving the linear block of IMN-Hybrid its own, weaker L1 penalty ($\lambda = 0.1$) does not change accuracy (+0.0003, $p = 0.339$). The main-stage hypothesis that the strong prior penalty suppressed a useful linear term is not supported.

Table 8: Tier 2: test AUROC (mean$_{\pm sd}$ over 3 seeds), default hyper-parameters with the step cap. Layout as in Table 3.

Dataset	LR	DT	RF	GBDT	TabResNet	IMN
mfeat-factors	0.997$_{\pm 0.001}$	0.931$_{\pm 0.017}$	0.998$_{\pm 0.001}$	0.998$_{\pm 0.000}$	**0.999$_{\pm 0.000}$**	0.998$_{\pm 0.000}$
KDDCup09_appetency	0.770$_{\pm 0.009}$	0.527$_{\pm 0.011}$	0.791$_{\pm 0.006}$	**0.807$_{\pm 0.003}$**	0.744$_{\pm 0.011}$	0.759$_{\pm 0.003}$
airlines	0.680$_{\pm 0.002}$	0.589$_{\pm 0.001}$	0.693$_{\pm 0.003}$	**0.704$_{\pm 0.002}$**	0.677$_{\pm 0.003}$	0.676$_{\pm 0.004}$
bank-marketing	0.904$_{\pm 0.003}$	0.702$_{\pm 0.003}$	0.928$_{\pm 0.004}$	**0.930$_{\pm 0.001}$**	0.904$_{\pm 0.002}$	0.903$_{\pm 0.002}$
cnae-9	**0.995$_{\pm 0.003}$**	0.911$_{\pm 0.004}$	0.993$_{\pm 0.002}$	0.970$_{\pm 0.006}$	0.992$_{\pm 0.004}$	0.983$_{\pm 0.004}$
nomao	0.986$_{\pm 0.001}$	0.928$_{\pm 0.004}$	0.993$_{\pm 0.001}$	**0.993$_{\pm 0.001}$**	0.986$_{\pm 0.000}$	0.983$_{\pm 0.001}$
Amazon_employee_access	0.845$_{\pm 0.012}$	0.645$_{\pm 0.023}$	0.849$_{\pm 0.020}$	**0.857$_{\pm 0.018}$**	0.848$_{\pm 0.010}$	0.847$_{\pm 0.010}$
higgs	0.684$_{\pm 0.002}$	0.624$_{\pm 0.001}$	0.799$_{\pm 0.005}$	**0.801$_{\pm 0.004}$**	0.721$_{\pm 0.006}$	0.705$_{\pm 0.004}$
numerai28.6	**0.529$_{\pm 0.002}$**	0.503$_{\pm 0.003}$	0.516$_{\pm 0.006}$	0.523$_{\pm 0.001}$	0.502$_{\pm 0.009}$	0.521$_{\pm 0.007}$
shuttle	0.961$_{\pm 0.005}$	0.956$_{\pm 0.043}$	**1.000$_{\pm 0.000}$**	0.578$_{\pm 0.028}$	0.947$_{\pm 0.030}$	0.906$_{\pm 0.031}$
Fashion-MNIST	0.981$_{\pm 0.000}$	0.883$_{\pm 0.002}$	0.991$_{\pm 0.000}$	**0.993$_{\pm 0.000}$**	0.985$_{\pm 0.001}$	0.984$_{\pm 0.000}$
APSFailure	0.961$_{\pm 0.002}$	0.853$_{\pm 0.007}$	0.991$_{\pm 0.002}$	**0.992$_{\pm 0.002}$**	0.987$_{\pm 0.001}$	0.980$_{\pm 0.006}$
christine	0.718$_{\pm 0.009}$	0.636$_{\pm 0.011}$	0.798$_{\pm 0.018}$	**0.811$_{\pm 0.011}$**	0.737$_{\pm 0.008}$	0.767$_{\pm 0.007}$
MiniBooNE	0.937$_{\pm 0.001}$	0.866$_{\pm 0.003}$	0.980$_{\pm 0.001}$	**0.983$_{\pm 0.001}$**	0.925$_{\pm 0.006}$	0.907$_{\pm 0.002}$
riccardo	0.995$_{\pm 0.002}$	0.837$_{\pm 0.009}$	0.998$_{\pm 0.000}$	**1.000$_{\pm 0.000}$**	0.998$_{\pm 0.001}$	0.998$_{\pm 0.001}$
dilbert	0.994$_{\pm 0.000}$	0.868$_{\pm 0.004}$	0.998$_{\pm 0.000}$	**1.000$_{\pm 0.000}$**	0.996$_{\pm 0.000}$	0.996$_{\pm 0.000}$
fabert	0.873$_{\pm 0.008}$	0.745$_{\pm 0.009}$	**0.920$_{\pm 0.005}$**	0.907$_{\pm 0.005}$	0.901$_{\pm 0.005}$	0.899$_{\pm 0.006}$
volkert	0.889$_{\pm 0.002}$	0.693$_{\pm 0.002}$	0.934$_{\pm 0.001}$	**0.937$_{\pm 0.001}$**	0.878$_{\pm 0.003}$	0.891$_{\pm 0.003}$
jannis	0.813$_{\pm 0.002}$	0.607$_{\pm 0.004}$	0.861$_{\pm 0.001}$	**0.868$_{\pm 0.003}$**	0.804$_{\pm 0.002}$	0.816$_{\pm 0.005}$
helena	0.859$_{\pm 0.003}$	0.549$_{\pm 0.002}$	**0.864$_{\pm 0.000}$**	0.658$_{\pm 0.007}$	0.849$_{\pm 0.003}$	0.857$_{\pm 0.003}$

Dataset	IMN	TM	Clause-LR	IMN-TM	IMN-TM (rand.)	IMN-Hybrid	TM-Gated	Paper IMN
mfeat-factors	0.998$_{\pm 0.000}$	0.997$_{\pm 0.002}$	0.999$_{\pm 0.001}$	0.998$_{\pm 0.001}$	**0.999$_{\pm 0.001}$**	0.998$_{\pm 0.001}$	0.993$_{\pm 0.001}$	0.999
KDDCup09_appetency	0.759$_{\pm 0.003}$	0.751$_{\pm 0.008}$	**0.813$_{\pm 0.002}$**	0.806$_{\pm 0.002}$	0.761$_{\pm 0.034}$	0.808$_{\pm 0.002}$	0.757$_{\pm 0.020}$	0.816
airlines	0.676$_{\pm 0.004}$	0.658$_{\pm 0.005}$	**0.682$_{\pm 0.002}$**	0.672$_{\pm 0.007}$	0.671$_{\pm 0.014}$	0.673$_{\pm 0.002}$	0.652$_{\pm 0.011}$	0.695
bank-marketing	0.903$_{\pm 0.002}$	0.890$_{\pm 0.007}$	0.904$_{\pm 0.005}$	0.901$_{\pm 0.005}$	0.901$_{\pm 0.004}$	0.903$_{\pm 0.005}$	**0.909$_{\pm 0.004}$**	0.930
cnae-9	0.983$_{\pm 0.004}$	0.987$_{\pm 0.005}$	0.996$_{\pm 0.001}$	**0.996$_{\pm 0.002}$**	0.975$_{\pm 0.006}$	0.987$_{\pm 0.005}$	0.967$_{\pm 0.006}$	0.994
nomao	0.983$_{\pm 0.001}$	0.985$_{\pm 0.001}$	0.989$_{\pm 0.000}$	0.989$_{\pm 0.001}$	0.987$_{\pm 0.001}$	0.989$_{\pm 0.001}$	**0.990$_{\pm 0.000}$**	0.993
Amazon_employee_access	0.847$_{\pm 0.010}$	0.809$_{\pm 0.014}$	**0.850$_{\pm 0.013}$**	0.844$_{\pm 0.016}$	0.848$_{\pm 0.015}$	0.844$_{\pm 0.016}$	0.843$_{\pm 0.016}$	0.858
higgs	0.705$_{\pm 0.004}$	0.754$_{\pm 0.004}$	**0.772$_{\pm 0.005}$**	0.765$_{\pm 0.005}$	0.706$_{\pm 0.002}$	0.769$_{\pm 0.005}$	0.752$_{\pm 0.005}$	0.823
numerai28.6	**0.521$_{\pm 0.007}$**	0.511$_{\pm 0.006}$	0.520$_{\pm 0.005}$	0.509$_{\pm 0.004}$	0.504$_{\pm 0.011}$	0.510$_{\pm 0.004}$	0.509$_{\pm 0.006}$	0.530
shuttle	0.906$_{\pm 0.031}$	0.952$_{\pm 0.040}$	0.997$_{\pm 0.001}$	0.998$_{\pm 0.001}$	0.997$_{\pm 0.001}$	**0.999$_{\pm 0.001}$**	0.895$_{\pm 0.102}$	0.993
Fashion-MNIST	0.984$_{\pm 0.000}$	0.982$_{\pm 0.000}$	0.990$_{\pm 0.000}$	**0.990$_{\pm 0.001}$**	0.985$_{\pm 0.000}$	0.989$_{\pm 0.000}$	0.932$_{\pm 0.004}$	0.992
APSFailure	0.980$_{\pm 0.006}$	0.981$_{\pm 0.005}$	0.988$_{\pm 0.002}$	0.988$_{\pm 0.003}$	0.983$_{\pm 0.001}$	**0.989$_{\pm 0.002}$**	0.847$_{\pm 0.032}$	0.980
christine	0.767$_{\pm 0.007}$	0.772$_{\pm 0.009}$	0.768$_{\pm 0.007}$	**0.776$_{\pm 0.009}$**	0.745$_{\pm 0.002}$	0.771$_{\pm 0.007}$	0.775$_{\pm 0.009}$	0.775
MiniBooNE	0.907$_{\pm 0.002}$	0.977$_{\pm 0.001}$	0.980$_{\pm 0.001}$	0.980$_{\pm 0.001}$	0.942$_{\pm 0.012}$	0.980$_{\pm 0.001}$	**0.981$_{\pm 0.001}$**	0.984
riccardo	0.998$_{\pm 0.001}$	0.998$_{\pm 0.001}$	0.999$_{\pm 0.001}$	0.999$_{\pm 0.001}$	0.998$_{\pm 0.001}$	**0.999$_{\pm 0.000}$**	0.998$_{\pm 0.001}$	0.998
dilbert	0.996$_{\pm 0.000}$	0.995$_{\pm 0.001}$	0.998$_{\pm 0.000}$	0.998$_{\pm 0.000}$	0.997$_{\pm 0.001}$	**0.999$_{\pm 0.000}$**	0.994$_{\pm 0.001}$	1.000
fabert	0.899$_{\pm 0.006}$	0.535$_{\pm 0.008}$	0.581$_{\pm 0.019}$	0.899$_{\pm 0.007}$	**0.900$_{\pm 0.007}$**	0.898$_{\pm 0.011}$	0.866$_{\pm 0.019}$	0.902
volkert	0.891$_{\pm 0.003}$	0.877$_{\pm 0.004}$	0.915$_{\pm 0.002}$	0.915$_{\pm 0.002}$	0.902$_{\pm 0.001}$	**0.919$_{\pm 0.002}$**	0.824$_{\pm 0.004}$	0.943
jannis	0.816$_{\pm 0.005}$	0.804$_{\pm 0.001}$	0.847$_{\pm 0.001}$	0.847$_{\pm 0.001}$	0.809$_{\pm 0.002}$	**0.849$_{\pm 0.000}$**	0.791$_{\pm 0.001}$	0.856
helena	0.857$_{\pm 0.003}$	0.780$_{\pm 0.002}$	0.871$_{\pm 0.001}$	0.870$_{\pm 0.002}$	0.865$_{\pm 0.001}$	**0.871$_{\pm 0.002}$**	0.773$_{\pm 0.002}$	0.865

Table 9: Preregistered comparisons on Tier 2 (20 datasets). *: Holm-corrected $p < 0.05$.

Q	Comparison A vs. B	mean $A-B$	95% bootstrap CI	W/L	p	p_{Holm}
Q1	IMN-TM vs. TM	+0.0372	[+0.0126, +0.0772]	19/1	9.5×10^{-6}	9.5×10^{-5}*
Q2	IMN-TM vs. IMN	+0.0183	[+0.0072, +0.0314]	15/5	0.004	0.030*
Q3	IMN-Hybrid vs. IMN	+0.0185	[+0.0071, +0.0319]	15/5	0.003	0.025*
Q3	IMN-Hybrid vs. IMN-TM	+0.0002	[-0.0011, +0.0013]	14/6	0.064	0.350
Q4	TM-Gated vs. LR	-0.0163	[-0.0354, +0.0024]	6/14	0.090	0.359
Q4	TM-Gated vs. TM	+0.0025	[-0.0271, +0.0442]	7/13	0.294	0.883
Q4	TM-Gated vs. IMN	-0.0164	[-0.0359, +0.0019]	5/15	0.058	0.350
Q5	IMN-TM vs. Clause-LR	+0.0141	[-0.0035, +0.0470]	7/13	0.312	0.883
Q6	IMN-TM vs. IMN-TM (rand.)	+0.0133	[+0.0058, +0.0216]	16/4	5.9×10^{-4}	0.005*
R	IMN vs. TabResNet	-0.0003	[-0.0069, +0.0059]	7/13	0.648	0.883

Table 10: Synthetic attribution benchmark: mean over ρ and seeds, per family. GT corr.: correlation of the removal attributions with the true contributions. infid.: infidelity under small Gaussian perturbations. It penalises piecewise-constant models, whose output jumps, and is reported for completeness only.

	Gaussian linear			Non-linear additive			Piecewise constant		
Method	AUROC	GT corr. ↑	infid. ↓	AUROC	GT corr. ↑	infid. ↓	AUROC	GT corr. ↑	infid. ↓
LR	0.997	0.975	2.581	0.741	0.523	0.034	0.757	0.576	0.037
GBDT	0.990	0.934	1.026	0.984	0.612	0.960	0.990	0.638	4.512
TabResNet	0.997	0.972	0.617	0.970	0.656	0.221	0.847	0.607	0.060
IMN	0.997	0.843	0.245	0.968	0.541	0.188	0.848	0.570	0.057
TM	0.983	0.826	0.342	0.969	0.518	0.488	0.939	0.558	0.617
Clause-LR	0.987	0.939	1.441	0.976	0.578	1.883	0.947	0.625	1.732
IMN-TM	0.990	0.919	2.086	0.974	0.570	2.440	0.944	0.616	1.993
IMN-Hybrid	0.990	0.945	2.200	0.975	0.572	2.444	0.945	0.620	1.980
TM-Gated	0.989	0.736	0.827	0.978	0.400	1.366	0.948	0.229	1.698

Table 11: Ablations (Tier 1, seeds 0–2). Δ is the test AUROC change versus the reference arm on the same seeds; p is an uncorrected two-sided Wilcoxon test over datasets. The last column is local fidelity at $k = 10$.

Variant	reference	n	mean Δ	W/L	p	Δ vs. IMN	fid. k=10
TM, 50 clauses/class	TM	12	-0.0034	7/5	0.733	–	–
TM, 800 clauses/class	TM	12	+0.0036	9/3	0.077	–	–
IMN-TM on 50-clause TM	IMN-TM	12	+0.0004	6/6	0.850	-0.0006	0.708
IMN-TM on 800-clause TM	IMN-TM	12	-0.0012	8/4	0.622	-0.0021	0.741
IMN-TM without TM initialisation	IMN-TM	12	+0.0002	6/6	0.970	-0.0007	0.649
IMN-TM, generator input $[x, c(x)]$	IMN-TM	12	-0.0116	5/7	0.064	-0.0126	0.706
IMN-TM, random clauses (rate-matched)	IMN-TM	12	-0.0017	8/4	0.850	-0.0026	0.557
IMN-Hybrid, linear $\lambda = 0.1$	IMN-Hybrid	12	+0.0003	8/4	0.339	+0.0001	0.738

5 Limitations

- **Two-stage training.** The clause basis is fixed before the hypernetwork sees it: the TM's clauses are learned for its own fixed vote, not for per-example re-weighting. Random clauses match the learned ones in AUROC, so better bases are an open direction.
- **Explanation length.** Clause explanations are long, and an accuracy-only search makes them longer. The removal attribution of a clause model depends on the baseline.
- **Hyper-parameter selection.** Winners were selected on one validation split (seed 0). On small datasets this is noisy: the tuned IMN lost blood-transfusion (-0.073) against its own default.
- **Reproduction.** Our split differs from the paper's. Our default-HP IMN falls short of the paper's numbers on some datasets (phoneme (-0.068), jungle_chess (-0.061)), which flatters the TM arms there. Tuning closes these gaps.
- **Substitutions and caps.** CatBoost was replaced by HistGradientBoosting. Tier-2 runs use a step cap that binds on 15 of 20 datasets. It favours models with a good initialisation, so Tier-2 differences between IMN and the clause models partly measure training efficiency (Section 4.5). Three Tier-2 datasets could not be downloaded verifiably and are excluded.

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